

Network Investigation

Exercise: 1.4 | External Network Communication Detected

Objectives:

The primary objective is to conduct a thorough investigation into a suspected external network communication Incident with the goal of identifying the source, scope, and impact of the attack.

Incident Details:

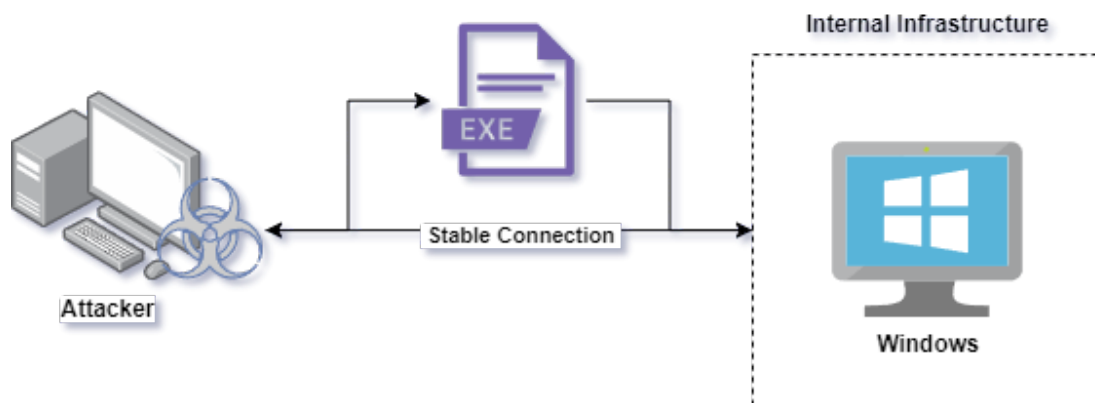
Investigate the suspicious external communication activity detected by the network monitoring systems. Investigate the PCAP to identify the targeted IP, source, and potential impact.

PCAP File Name:

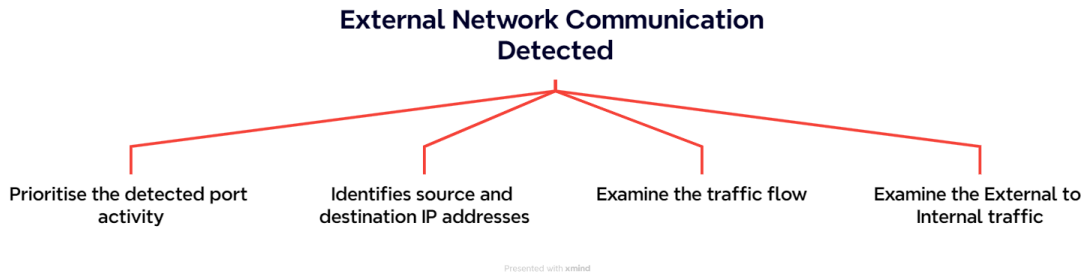
Targeted Port:

About RFI:

External network communication refers to the interaction and data exchange between a local network and external networks, when a attacker successfully deploy the payload over the host machine it will establish a stable connection between the attacker and the victim



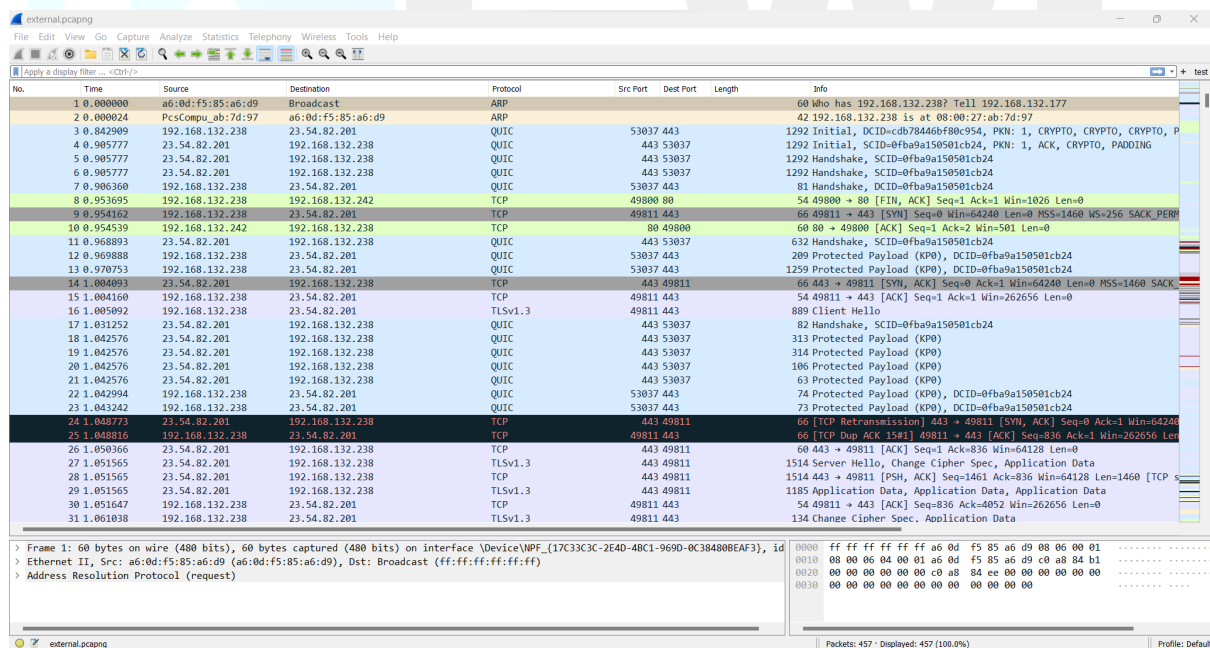
Investigation Mindmap:



Step:01

The first step of the investigation begins by analysing the given network PCAP. We have provided you with the suspected PCAP file which has been extracted from one of the compromised host machines.

Download the given PCAP dump on your own host machine for further investigation.



No.	Time	Source	Destination	Protocol	Src Port	Dest Port	Length	Info
1	0.000000	a6:0d:f5:85:a6:d9	Broadcast	ARP				60 who has 192.168.132.238? Tell 192.168.132.177
2	0.000024	PcsCompu_ab:7d:97	a6:0d:f5:85:a6:d9	ARP				42 192.168.132.238 is at 08:00:27:ab:7d:97
3	0.842909	192.168.132.238	23.54.82.201	QUIC	53037	443		1292 Initial, DCID=cdb78446bf80c954, PKN: 1, CRYPTO, CRYPTO, CRYPTO, P
4	0.905777	23.54.82.201	192.168.132.238	QUIC	443	53037		1292 Initial, SCID=0fba9a150501cb24, PKN: 1, ACK, CRYPTO, PADDING
5	0.905777	23.54.82.201	192.168.132.238	QUIC	443	53037		1292 Handshake, SCID=0fba9a150501cb24
6	0.905777	23.54.82.201	192.168.132.238	QUIC	443	53037		1292 Handshake, SCID=0fba9a150501cb24
7	0.953660	192.168.132.238	23.54.82.201	QUIC	53037	443		81 Handshake, DCID=0fba9a150501cb24
8	0.953660	192.168.132.238	192.168.132.242	TCP	49800	80		54 49800 → 80 [FIN, ACK] Seq=1 Ack=1 Win=1026 Len=0
9	0.954162	192.168.132.238	23.54.82.201	TCP	49811	443		66 49811 → 443 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 WS=256 SACK_PERM
10	0.954539	192.168.132.242	192.168.132.238	TCP	80	49800		60 80 → 49800 [ACK] Seq=1 Ack=2 Win=501 Len=0
11	0.968893	23.54.82.201	192.168.132.238	QUIC	443	53037		632 Handshake, SCID=0fba9a150501cb24
12	0.969008	192.168.132.238	23.54.82.201	QUIC	53037	443		289 Protected Payload (KP0), DCID=0fba9a150501cb24
13	0.970753	192.168.132.238	23.54.82.201	QUIC	53037	443		1259 Protected Payload (KP0), DCID=0fba9a150501cb24
14	1.004093	23.54.82.201	192.168.132.238	TCP	443	49811		66 443 → 49811 [SYN, ACK] Seq=0 Ack=1 Win=64240 Len=0 MSS=1460 SACK
15	1.004160	192.168.132.238	23.54.82.201	TCP	49811	443		54 49811 → 443 [ACK] Seq=1 Ack=1 Win=262656 Len=0
16	1.005092	192.168.132.238	23.54.82.201	TLSv1.3	49811	443		809 Client Hello
17	1.031252	23.54.82.201	192.168.132.238	QUIC	443	53037		82 Handshake, SCID=0fba9a150501cb24
18	1.042576	23.54.82.201	192.168.132.238	QUIC	443	53037		313 Protected Payload (KP0)
19	1.042576	23.54.82.201	192.168.132.238	QUIC	443	53037		314 Protected Payload (KP0)
20	1.042576	23.54.82.201	192.168.132.238	QUIC	443	53037		106 Protected Payload (KP0)
21	1.042576	23.54.82.201	192.168.132.238	QUIC	443	53037		63 Protected Payload (KP0)
22	1.042994	192.168.132.238	23.54.82.201	QUIC	53037	443		74 Protected Payload (KP0), DCID=0fba9a150501cb24
23	1.043242	192.168.132.238	23.54.82.201	QUIC	53037	443		73 Protected Payload (KP0), DCID=0fba9a150501cb24
24	1.048773	23.54.82.201	192.168.132.238	TCP	443	49811		66 [TCP Retransmission] 443 → 49811 [SYN, ACK] Seq=0 Ack=1 Win=64240
25	1.048816	192.168.132.238	23.54.82.201	TCP	49811	443		66 [TCP Dup ACK 15#1] 49811 → 443 [ACK] Seq=836 Ack=1 Win=262656 Len
26	1.050366	23.54.82.201	192.168.132.238	TCP	443	49811		60 443 → 49811 [ACK] Seq=1 Ack=836 Win=64128 Len=0
27	1.051565	23.54.82.201	192.168.132.238	TLSv1.3	443	49811		1514 Server Hello, Change Cipher Spec, Application Data
28	1.051565	23.54.82.201	192.168.132.238	TCP	443	49811		1514 443 → 49811 [PSH, ACK] Seq=1461 Ack=836 Win=64128 Len=1460 [TCP
29	1.051565	23.54.82.201	192.168.132.238	TLSv1.3	443	49811		1185 Application Data, Application Data
30	1.051647	192.168.132.238	23.54.82.201	TCP	49811	443		54 49811 → 443 [ACK] Seq=836 Ack=4052 Win=262656 Len=0
31	1.061038	192.168.132.238	23.54.82.201	TLSv1.3	49811	443		134 Change Cipher Spec, Application Data

> Frame 1: 60 bytes on wire (480 bits), 60 bytes captured (480 bits) on interface \Device\NPF_{17C33C3C-2E4D-4BC1-9690-0C384808EAF3}, id 0000
 > Ethernet II, Src: a6:0d:f5:85:a6:d9 (a6:0d:f5:85:a6:d9), Dst: Broadcast (ff:ff:ff:ff:ff:ff)
 > Address Resolution Protocol (request)

Packets: 457 · Displayed: 457 (100.0%) Profile: Default

Wireshark is a popular open-source network protocol analyzer that allows users to capture and inspect the data travelling back and forth on a network in real time

Step:03

Investigating external communication involves careful steps to ensure security and determine the nature and potential threats associated with the network logs.

1. Prioritise the detected port activity
2. Identifies source and destination IP addresses
3. Examine the traffic flow
4. Examine the External to Internal traffic

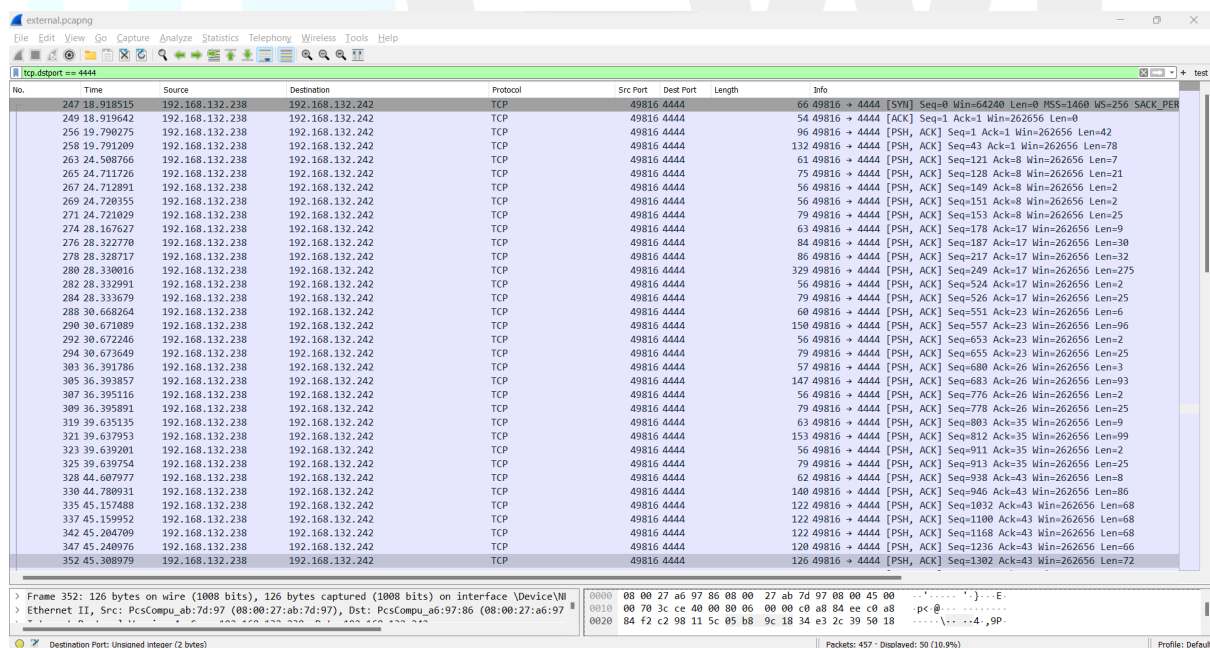
Prioritise the detected port activity

Identifying and extracting particular logs which are associated with the identified activity

Use the following command to obtain the web-access log.

Unset

```
tcp.dstport == 4444
```



No.	Time	Source	Destination	Protocol	Src Port	Dest Port	Length	Info
247	18.918515	192.168.132.238	192.168.132.242	TCP	49816	4444	66	49816 → 4444 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 WS=256 SACK_PER
249	18.919642	192.168.132.238	192.168.132.242	TCP	49816	4444	54	49816 → 4444 [ACK] Seq=1 Ack=1 Win=262656 Len=0
256	19.790275	192.168.132.238	192.168.132.242	TCP	49816	4444	96	49816 → 4444 [PSH, ACK] Seq=1 Ack=1 Win=262656 Len=42
258	19.791200	192.168.132.238	192.168.132.242	TCP	49816	4444	132	49816 → 4444 [PSH, ACK] Seq=43 Ack=1 Win=262656 Len=78
263	24.508766	192.168.132.238	192.168.132.242	TCP	49816	4444	61	49816 → 4444 [PSH, ACK] Seq=121 Ack=8 Win=262656 Len=7
265	24.711726	192.168.132.238	192.168.132.242	TCP	49816	4444	75	49816 → 4444 [PSH, ACK] Seq=128 Ack=8 Win=262656 Len=21
267	24.712891	192.168.132.238	192.168.132.242	TCP	49816	4444	56	49816 → 4444 [PSH, ACK] Seq=149 Ack=8 Win=262656 Len=2
269	24.720355	192.168.132.238	192.168.132.242	TCP	49816	4444	56	49816 → 4444 [PSH, ACK] Seq=151 Ack=8 Win=262656 Len=2
271	24.721029	192.168.132.238	192.168.132.242	TCP	49816	4444	79	49816 → 4444 [PSH, ACK] Seq=153 Ack=8 Win=262656 Len=25
274	28.167627	192.168.132.238	192.168.132.242	TCP	49816	4444	63	49816 → 4444 [PSH, ACK] Seq=178 Ack=17 Win=262656 Len=9
276	28.322770	192.168.132.238	192.168.132.242	TCP	49816	4444	84	49816 → 4444 [PSH, ACK] Seq=187 Ack=17 Win=262656 Len=30
278	28.328717	192.168.132.238	192.168.132.242	TCP	49816	4444	86	49816 → 4444 [PSH, ACK] Seq=217 Ack=17 Win=262656 Len=32
280	28.330016	192.168.132.238	192.168.132.242	TCP	49816	4444	329	49816 → 4444 [PSH, ACK] Seq=249 Ack=17 Win=262656 Len=275
282	28.332991	192.168.132.238	192.168.132.242	TCP	49816	4444	56	49816 → 4444 [PSH, ACK] Seq=524 Ack=17 Win=262656 Len=2
284	28.333679	192.168.132.238	192.168.132.242	TCP	49816	4444	79	49816 → 4444 [PSH, ACK] Seq=526 Ack=17 Win=262656 Len=25
288	30.668264	192.168.132.238	192.168.132.242	TCP	49816	4444	60	49816 → 4444 [PSH, ACK] Seq=551 Ack=23 Win=262656 Len=6
290	30.671089	192.168.132.238	192.168.132.242	TCP	49816	4444	150	49816 → 4444 [PSH, ACK] Seq=557 Ack=23 Win=262656 Len=96
292	30.672246	192.168.132.238	192.168.132.242	TCP	49816	4444	56	49816 → 4444 [PSH, ACK] Seq=653 Ack=23 Win=262656 Len=2
294	30.673649	192.168.132.238	192.168.132.242	TCP	49816	4444	79	49816 → 4444 [PSH, ACK] Seq=655 Ack=23 Win=262656 Len=25
303	36.391786	192.168.132.238	192.168.132.242	TCP	49816	4444	57	49816 → 4444 [PSH, ACK] Seq=680 Ack=26 Win=262656 Len=3
305	36.393857	192.168.132.238	192.168.132.242	TCP	49816	4444	147	49816 → 4444 [PSH, ACK] Seq=683 Ack=26 Win=262656 Len=93
307	36.395116	192.168.132.238	192.168.132.242	TCP	49816	4444	56	49816 → 4444 [PSH, ACK] Seq=776 Ack=26 Win=262656 Len=2
309	36.395891	192.168.132.238	192.168.132.242	TCP	49816	4444	79	49816 → 4444 [PSH, ACK] Seq=778 Ack=26 Win=262656 Len=25
319	39.635135	192.168.132.238	192.168.132.242	TCP	49816	4444	63	49816 → 4444 [PSH, ACK] Seq=803 Ack=35 Win=262656 Len=9
321	39.637953	192.168.132.238	192.168.132.242	TCP	49816	4444	153	49816 → 4444 [PSH, ACK] Seq=812 Ack=35 Win=262656 Len=99
323	39.639201	192.168.132.238	192.168.132.242	TCP	49816	4444	56	49816 → 4444 [PSH, ACK] Seq=911 Ack=35 Win=262656 Len=2
325	39.639754	192.168.132.238	192.168.132.242	TCP	49816	4444	79	49816 → 4444 [PSH, ACK] Seq=913 Ack=35 Win=262656 Len=25
328	44.607977	192.168.132.238	192.168.132.242	TCP	49816	4444	62	49816 → 4444 [PSH, ACK] Seq=938 Ack=43 Win=262656 Len=8
330	44.780931	192.168.132.238	192.168.132.242	TCP	49816	4444	140	49816 → 4444 [PSH, ACK] Seq=946 Ack=43 Win=262656 Len=86
335	45.157488	192.168.132.238	192.168.132.242	TCP	49816	4444	122	49816 → 4444 [PSH, ACK] Seq=1032 Ack=43 Win=262656 Len=68
337	45.159952	192.168.132.238	192.168.132.242	TCP	49816	4444	122	49816 → 4444 [PSH, ACK] Seq=1100 Ack=43 Win=262656 Len=68
342	45.204709	192.168.132.238	192.168.132.242	TCP	49816	4444	122	49816 → 4444 [PSH, ACK] Seq=1168 Ack=43 Win=262656 Len=68
347	45.240976	192.168.132.238	192.168.132.242	TCP	49816	4444	120	49816 → 4444 [PSH, ACK] Seq=1236 Ack=43 Win=262656 Len=66
352	45.308979	192.168.132.238	192.168.132.242	TCP	49816	4444	126	49816 → 4444 [PSH, ACK] Seq=1302 Ack=43 Win=262656 Len=72

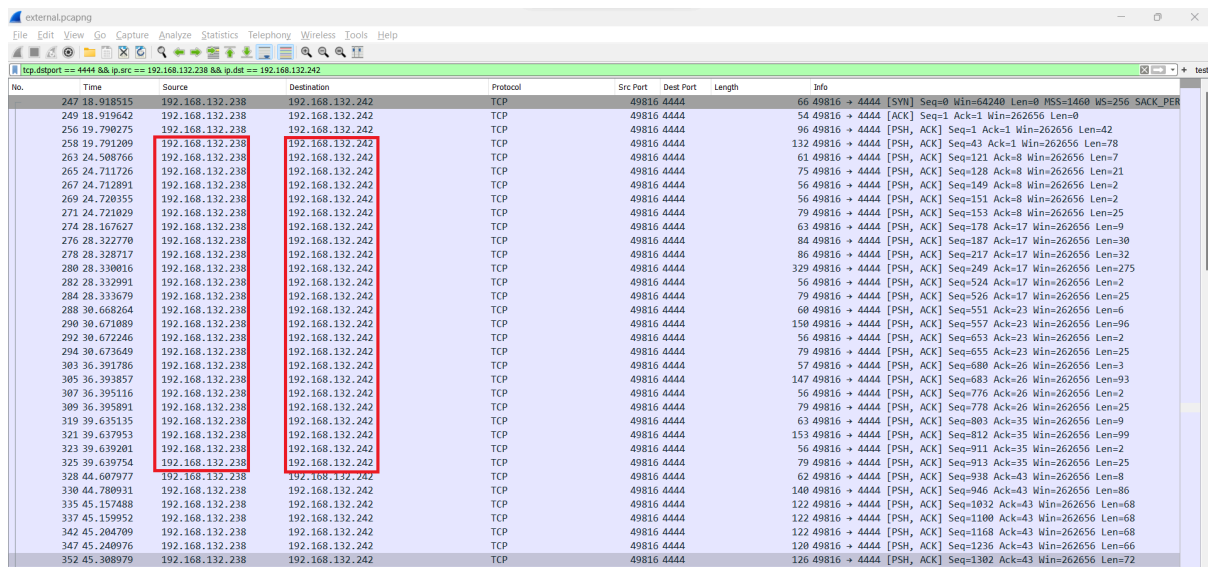
Decoders are the key components which are responsible for parsing raw logs into a structured format that can be easily analysed and understood by the system.

Identifies source and destination IP addresses

The next step is to identify and retrieve the IP activity which are associated with the detected port activity, from the above result we identified that the src 192.168.132.238 → 192.168.132.242 over destination port 4444

Unset

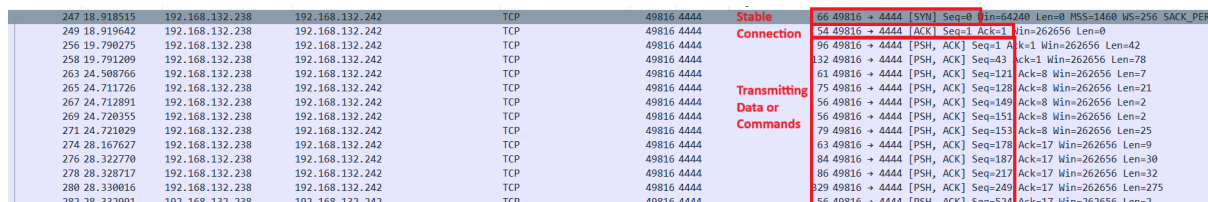
```
tcp.dstport == 4444 && ip.src == 192.168.132.238 && ip.dst == 192.168.132.242
```



No.	Time	Source	Destination	Protocol	Src Port	Dest Port	Length	Info
247	18.918515	192.168.132.238	192.168.132.242	TCP	49816	4444	66	49816 → 4444 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 WS=256 SACK_PER
249	18.919642	192.168.132.238	192.168.132.242	TCP	49816	4444	54	49816 → 4444 [ACK] Seq=1 Ack=1 Win=262656 Len=0
256	19.790275	192.168.132.238	192.168.132.242	TCP	49816	4444	96	49816 → 4444 [PSH, ACK] Seq=1 Ack=1 Win=262656 Len=42
258	19.791209	192.168.132.238	192.168.132.242	TCP	49816	4444	132	49816 → 4444 [PSH, ACK] Seq=43 Ack=1 Win=262656 Len=78
263	24.508766	192.168.132.238	192.168.132.242	TCP	49816	4444	61	49816 → 4444 [PSH, ACK] Seq=121 Ack=8 Win=262656 Len=7
265	24.711726	192.168.132.238	192.168.132.242	TCP	49816	4444	75	49816 → 4444 [PSH, ACK] Seq=128 Ack=8 Win=262656 Len=21
267	24.712091	192.168.132.238	192.168.132.242	TCP	49816	4444	56	49816 → 4444 [PSH, ACK] Seq=149 Ack=8 Win=262656 Len=2
269	24.720355	192.168.132.238	192.168.132.242	TCP	49816	4444	56	49816 → 4444 [PSH, ACK] Seq=151 Ack=8 Win=262656 Len=2
271	24.721029	192.168.132.238	192.168.132.242	TCP	49816	4444	79	49816 → 4444 [PSH, ACK] Seq=153 Ack=8 Win=262656 Len=25
274	28.167627	192.168.132.238	192.168.132.242	TCP	49816	4444	63	49816 → 4444 [PSH, ACK] Seq=178 Ack=17 Win=262656 Len=9
276	28.322770	192.168.132.238	192.168.132.242	TCP	49816	4444	84	49816 → 4444 [PSH, ACK] Seq=187 Ack=17 Win=262656 Len=30
278	28.328717	192.168.132.238	192.168.132.242	TCP	49816	4444	86	49816 → 4444 [PSH, ACK] Seq=217 Ack=17 Win=262656 Len=32
280	28.330016	192.168.132.238	192.168.132.242	TCP	49816	4444	329	49816 → 4444 [PSH, ACK] Seq=249 Ack=17 Win=262656 Len=275
282	28.332991	192.168.132.238	192.168.132.242	TCP	49816	4444	56	49816 → 4444 [PSH, ACK] Seq=524 Ack=17 Win=262656 Len=2
284	28.333679	192.168.132.238	192.168.132.242	TCP	49816	4444	79	49816 → 4444 [PSH, ACK] Seq=526 Ack=17 Win=262656 Len=25
288	30.668264	192.168.132.238	192.168.132.242	TCP	49816	4444	60	49816 → 4444 [PSH, ACK] Seq=551 Ack=23 Win=262656 Len=6
290	30.671089	192.168.132.238	192.168.132.242	TCP	49816	4444	150	49816 → 4444 [PSH, ACK] Seq=557 Ack=23 Win=262656 Len=96
292	30.672246	192.168.132.238	192.168.132.242	TCP	49816	4444	56	49816 → 4444 [PSH, ACK] Seq=653 Ack=23 Win=262656 Len=2
294	30.673649	192.168.132.238	192.168.132.242	TCP	49816	4444	79	49816 → 4444 [PSH, ACK] Seq=655 Ack=23 Win=262656 Len=25
303	36.391786	192.168.132.238	192.168.132.242	TCP	49816	4444	57	49816 → 4444 [PSH, ACK] Seq=680 Ack=26 Win=262656 Len=3
305	36.393857	192.168.132.238	192.168.132.242	TCP	49816	4444	147	49816 → 4444 [PSH, ACK] Seq=683 Ack=26 Win=262656 Len=93
307	36.395116	192.168.132.238	192.168.132.242	TCP	49816	4444	56	49816 → 4444 [PSH, ACK] Seq=776 Ack=26 Win=262656 Len=2
309	36.395891	192.168.132.238	192.168.132.242	TCP	49816	4444	79	49816 → 4444 [PSH, ACK] Seq=778 Ack=26 Win=262656 Len=25
319	39.635135	192.168.132.238	192.168.132.242	TCP	49816	4444	63	49816 → 4444 [PSH, ACK] Seq=803 Ack=35 Win=262656 Len=9
321	39.637953	192.168.132.238	192.168.132.242	TCP	49816	4444	153	49816 → 4444 [PSH, ACK] Seq=812 Ack=35 Win=262656 Len=99
323	39.639201	192.168.132.238	192.168.132.242	TCP	49816	4444	56	49816 → 4444 [PSH, ACK] Seq=911 Ack=35 Win=262656 Len=2
325	39.639754	192.168.132.238	192.168.132.242	TCP	49816	4444	79	49816 → 4444 [PSH, ACK] Seq=913 Ack=35 Win=262656 Len=25
328	44.607977	192.168.132.238	192.168.132.242	TCP	49816	4444	62	49816 → 4444 [PSH, ACK] Seq=938 Ack=43 Win=262656 Len=8
330	44.700931	192.168.132.238	192.168.132.242	TCP	49816	4444	140	49816 → 4444 [PSH, ACK] Seq=946 Ack=43 Win=262656 Len=86
335	45.157488	192.168.132.238	192.168.132.242	TCP	49816	4444	122	49816 → 4444 [PSH, ACK] Seq=1032 Ack=43 Win=262656 Len=68
337	45.159552	192.168.132.238	192.168.132.242	TCP	49816	4444	122	49816 → 4444 [PSH, ACK] Seq=1100 Ack=43 Win=262656 Len=68
342	45.204709	192.168.132.238	192.168.132.242	TCP	49816	4444	122	49816 → 4444 [PSH, ACK] Seq=1168 Ack=43 Win=262656 Len=68
347	45.240976	192.168.132.238	192.168.132.242	TCP	49816	4444	120	49816 → 4444 [PSH, ACK] Seq=1236 Ack=43 Win=262656 Len=66
352	45.308979	192.168.132.238	192.168.132.242	TCP	49816	4444	126	49816 → 4444 [PSH, ACK] Seq=1302 Ack=43 Win=262656 Len=72

Examine the traffic flow

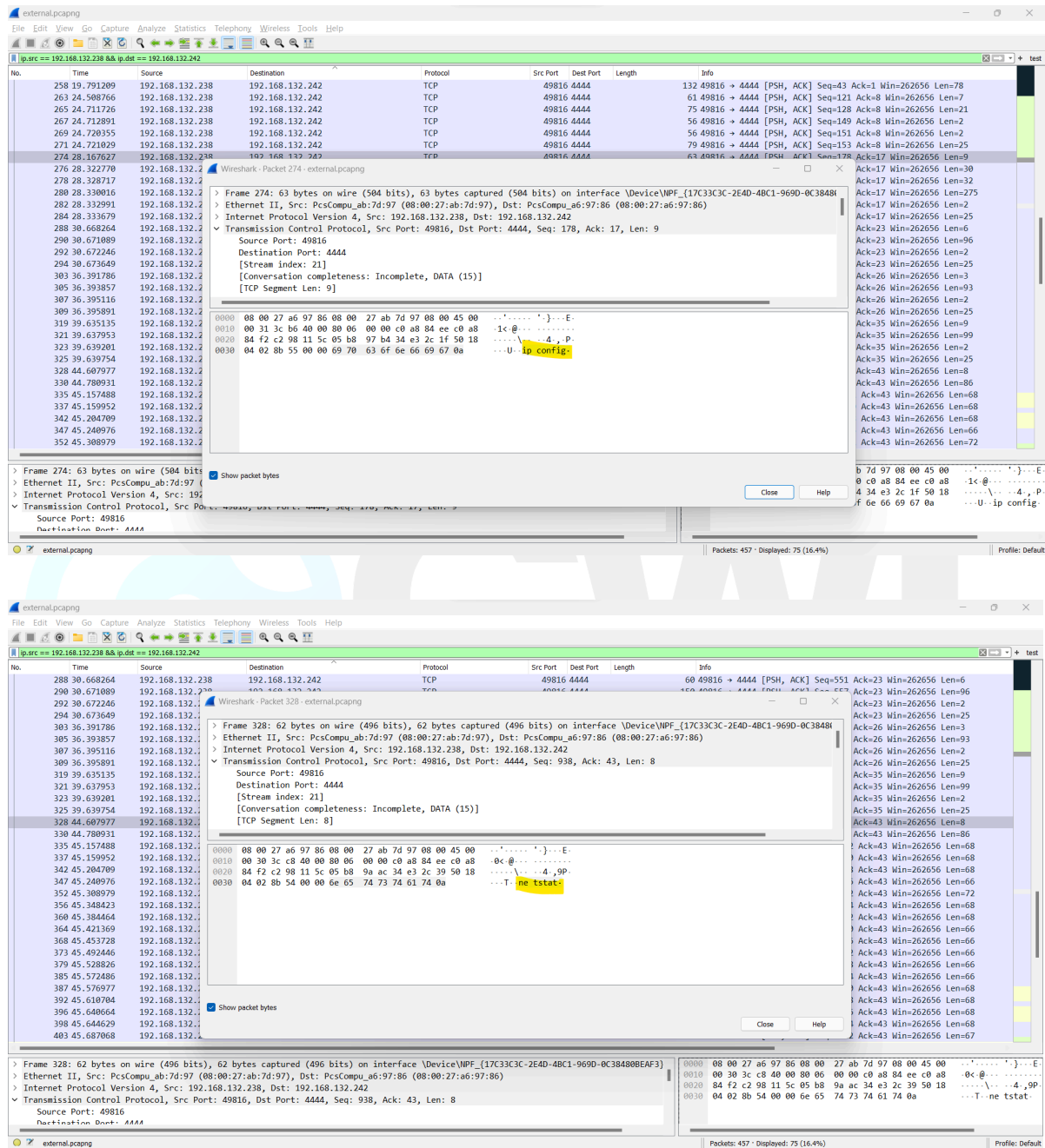
To examine the network traffic it is recommended to check the packet flag from the above observation result. We identified **SYN ACK** this pattern observed when a src dest established a stable communication, and **PSH** command generally observed when a src trying to transmit data or executing command.



No.	Time	Source	Destination	Protocol	Src Port	Dest Port	Length	Info
247	18.918515	192.168.132.238	192.168.132.242	TCP	49816	4444	66	49816 → 4444 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 WS=256 SACK_PER
249	18.919642	192.168.132.238	192.168.132.242	TCP	49816	4444	54	49816 → 4444 [ACK] Seq=1 Ack=1 Win=262656 Len=0
256	19.790275	192.168.132.238	192.168.132.242	TCP	49816	4444	96	49816 → 4444 [PSH, ACK] Seq=1 Ack=1 Win=262656 Len=42
258	19.791209	192.168.132.238	192.168.132.242	TCP	49816	4444	132	49816 → 4444 [PSH, ACK] Seq=43 Ack=1 Win=262656 Len=78
263	24.508766	192.168.132.238	192.168.132.242	TCP	49816	4444	61	49816 → 4444 [PSH, ACK] Seq=121 Ack=8 Win=262656 Len=7
265	24.711726	192.168.132.238	192.168.132.242	TCP	49816	4444	75	49816 → 4444 [PSH, ACK] Seq=128 Ack=8 Win=262656 Len=21
267	24.712091	192.168.132.238	192.168.132.242	TCP	49816	4444	56	49816 → 4444 [PSH, ACK] Seq=149 Ack=8 Win=262656 Len=2
269	24.720355	192.168.132.238	192.168.132.242	TCP	49816	4444	56	49816 → 4444 [PSH, ACK] Seq=151 Ack=8 Win=262656 Len=2
271	24.721029	192.168.132.238	192.168.132.242	TCP	49816	4444	79	49816 → 4444 [PSH, ACK] Seq=153 Ack=8 Win=262656 Len=25
274	28.167627	192.168.132.238	192.168.132.242	TCP	49816	4444	63	49816 → 4444 [PSH, ACK] Seq=178 Ack=17 Win=262656 Len=9
276	28.322770	192.168.132.238	192.168.132.242	TCP	49816	4444	84	49816 → 4444 [PSH, ACK] Seq=187 Ack=17 Win=262656 Len=30
278	28.328717	192.168.132.238	192.168.132.242	TCP	49816	4444	86	49816 → 4444 [PSH, ACK] Seq=217 Ack=17 Win=262656 Len=32
280	28.330016	192.168.132.238	192.168.132.242	TCP	49816	4444	329	49816 → 4444 [PSH, ACK] Seq=249 Ack=17 Win=262656 Len=275
282	28.332991	192.168.132.238	192.168.132.242	TCP	49816	4444	56	49816 → 4444 [PSH, ACK] Seq=524 Ack=17 Win=262656 Len=2

This particular pattern was observed when a victim got compromised and established a stable shell with the attacker machine.

On further checking the packet we identified multiple os command associated with the network activity



The image displays two screenshots of a Wireshark packet capture analysis. The first screenshot shows packet 274, a TCP segment from 192.168.132.238 to 192.168.132.242, port 49816 to 4444. The packet details pane shows the raw data, and the packet bytes pane shows the hex data. The second screenshot shows packet 328, a TCP segment from 192.168.132.238 to 192.168.132.242, port 49816 to 4444. The packet details pane shows the raw data, and the packet bytes pane shows the hex data. Both screenshots show the hex data being decoded into ASCII, revealing multiple 'ip config' and 'netstat' commands.

Packet 274:

Frame 274: 63 bytes on wire (504 bits), 63 bytes captured (504 bits) on interface \Device\NPF_{17C33C3C-2E4D-4BC1-969D-0C384808EA3F}

Ethernet II, Src: PcsCompu_ab:7d:97 (08:00:27:ab:7d:97), Dst: PcsCompu_a6:97:86 (08:00:27:a6:97:86)

Internet Protocol Version 4, Src: 192.168.132.238, Dst: 192.168.132.242

Transmission Control Protocol, Src Port: 49816, Dst Port: 4444, Seq: 178, Ack: 17, Len: 9

Source Port: 49816

Destination Port: 4444

[Stream index: 21]

[Conversation completeness: Incomplete, DATA (15)]

[TCP Segment Len: 9]

0000 08 00 27 a6 97 86 08 00 27 ab 7d 97 08 00 45 00E-
0010 00 31 3c b6 40 00 00 06 00 00 c0 a8 84 ee c0 a8 -1<@.....
0020 84 f2 c2 98 11 5c 05 b8 97 b4 34 e3 2c 1f 50 184.,P
0030 04 02 8b 55 00 00 69 70 63 6f 6e 66 69 67 0aU-ip config

Packet 328:

Frame 328: 62 bytes on wire (496 bits), 62 bytes captured (496 bits) on interface \Device\NPF_{17C33C3C-2E4D-4BC1-969D-0C384808EA3F}

Ethernet II, Src: PcsCompu_ab:7d:97 (08:00:27:ab:7d:97), Dst: PcsCompu_a6:97:86 (08:00:27:a6:97:86)

Internet Protocol Version 4, Src: 192.168.132.238, Dst: 192.168.132.242

Transmission Control Protocol, Src Port: 49816, Dst Port: 4444, Seq: 938, Ack: 43, Len: 8

Source Port: 49816

Destination Port: 4444

[Stream index: 21]

[Conversation completeness: Incomplete, DATA (15)]

[TCP Segment Len: 8]

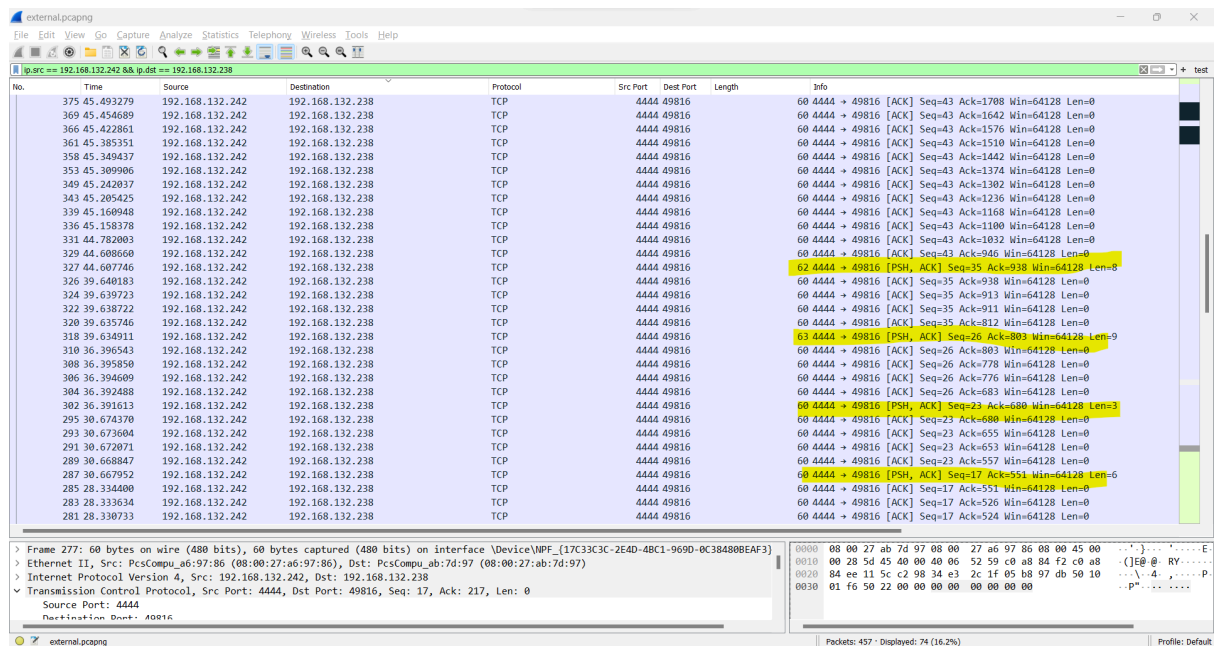
0000 08 00 27 a6 97 86 08 00 27 ab 7d 97 08 00 45 00E-
0010 00 30 3c c8 40 00 00 06 00 00 c0 a8 84 ee c0 a8 -0<@.....
0020 84 f2 c2 98 11 5c 05 b8 9a ac 34 e3 2c 39 50 184.,9P
0030 04 02 8b 54 00 00 6e 65 74 73 74 61 74 0aT-ne tstat

Examine the External to Internal traffic

After determining the suspicious attack pattern its recommend to check the reverse traffic vice verse, by examin the vice verse traffic its give a better visibility over the detected activity

Unset

```
ip.src == 192.168.132.242 && ip.dst == 192.168.132.238
```



The screenshot shows a Wireshark packet capture window titled 'external.pcapng'. The filter bar at the top is set to 'ip.src == 192.168.132.242 && ip.dst == 192.168.132.238'. The packet list shows 457 packets displayed, with 74 (16.2%) being the current view. The packet details pane shows the selected packet (No. 277) as an Ethernet II frame, followed by an Internet Protocol Version 4 packet, and a Transmission Control Protocol (TCP) packet. The packet bytes pane shows the raw data in hexadecimal and ASCII.

No.	Time	Source	Destination	Protocol	Src Port	Dest Port	Length	Info
375	45.492279	192.168.132.242	192.168.132.238	TCP	4444	49816	60	4444 → 49816 [ACK] Seq=43 Ack=1708 Win=64128 Len=0
369	45.454689	192.168.132.242	192.168.132.238	TCP	4444	49816	60	4444 → 49816 [ACK] Seq=43 Ack=1642 Win=64128 Len=0
366	45.422861	192.168.132.242	192.168.132.238	TCP	4444	49816	60	4444 → 49816 [ACK] Seq=43 Ack=1576 Win=64128 Len=0
361	45.385351	192.168.132.242	192.168.132.238	TCP	4444	49816	60	4444 → 49816 [ACK] Seq=43 Ack=1510 Win=64128 Len=0
358	45.349437	192.168.132.242	192.168.132.238	TCP	4444	49816	60	4444 → 49816 [ACK] Seq=43 Ack=1442 Win=64128 Len=0
353	45.309906	192.168.132.242	192.168.132.238	TCP	4444	49816	60	4444 → 49816 [ACK] Seq=43 Ack=1374 Win=64128 Len=0
349	45.242037	192.168.132.242	192.168.132.238	TCP	4444	49816	60	4444 → 49816 [ACK] Seq=43 Ack=1302 Win=64128 Len=0
343	45.205425	192.168.132.242	192.168.132.238	TCP	4444	49816	60	4444 → 49816 [ACK] Seq=43 Ack=1236 Win=64128 Len=0
339	45.160948	192.168.132.242	192.168.132.238	TCP	4444	49816	60	4444 → 49816 [ACK] Seq=43 Ack=1168 Win=64128 Len=0
336	45.158378	192.168.132.242	192.168.132.238	TCP	4444	49816	60	4444 → 49816 [ACK] Seq=43 Ack=1100 Win=64128 Len=0
331	44.782003	192.168.132.242	192.168.132.238	TCP	4444	49816	60	4444 → 49816 [ACK] Seq=43 Ack=1032 Win=64128 Len=0
329	44.608660	192.168.132.242	192.168.132.238	TCP	4444	49816	60	4444 → 49816 [ACK] Seq=43 Ack=966 Win=64128 Len=0
327	44.607746	192.168.132.242	192.168.132.238	TCP	4444	49816	60	4444 → 49816 [ACK] Seq=43 Ack=900 Win=64128 Len=0
326	39.640183	192.168.132.242	192.168.132.238	TCP	4444	49816	60	4444 → 49816 [ACK] Seq=35 Ack=938 Win=64128 Len=0
324	39.639723	192.168.132.242	192.168.132.238	TCP	4444	49816	60	4444 → 49816 [ACK] Seq=35 Ack=913 Win=64128 Len=0
322	39.638722	192.168.132.242	192.168.132.238	TCP	4444	49816	60	4444 → 49816 [ACK] Seq=35 Ack=911 Win=64128 Len=0
320	39.635746	192.168.132.242	192.168.132.238	TCP	4444	49816	60	4444 → 49816 [ACK] Seq=35 Ack=812 Win=64128 Len=0
318	39.634911	192.168.132.242	192.168.132.238	TCP	4444	49816	60	4444 → 49816 [ACK] Seq=35 Ack=803 Win=64128 Len=0
310	36.396543	192.168.132.242	192.168.132.238	TCP	4444	49816	60	4444 → 49816 [ACK] Seq=26 Ack=803 Win=64128 Len=0
308	36.395850	192.168.132.242	192.168.132.238	TCP	4444	49816	60	4444 → 49816 [ACK] Seq=26 Ack=778 Win=64128 Len=0
306	36.394609	192.168.132.242	192.168.132.238	TCP	4444	49816	60	4444 → 49816 [ACK] Seq=26 Ack=776 Win=64128 Len=0
304	36.392408	192.168.132.242	192.168.132.238	TCP	4444	49816	60	4444 → 49816 [ACK] Seq=26 Ack=683 Win=64128 Len=0
302	36.391613	192.168.132.242	192.168.132.238	TCP	4444	49816	60	4444 → 49816 [ACK] Seq=23 Ack=680 Win=64128 Len=0
295	30.674370	192.168.132.242	192.168.132.238	TCP	4444	49816	60	4444 → 49816 [ACK] Seq=23 Ack=655 Win=64128 Len=0
293	30.673604	192.168.132.242	192.168.132.238	TCP	4444	49816	60	4444 → 49816 [ACK] Seq=23 Ack=653 Win=64128 Len=0
291	30.672071	192.168.132.242	192.168.132.238	TCP	4444	49816	60	4444 → 49816 [ACK] Seq=23 Ack=557 Win=64128 Len=0
289	30.668847	192.168.132.242	192.168.132.238	TCP	4444	49816	60	4444 → 49816 [ACK] Seq=23 Ack=557 Win=64128 Len=0
287	30.667952	192.168.132.242	192.168.132.238	TCP	4444	49816	60	4444 → 49816 [ACK] Seq=17 Ack=551 Win=64128 Len=0
285	28.334080	192.168.132.242	192.168.132.238	TCP	4444	49816	60	4444 → 49816 [ACK] Seq=17 Ack=526 Win=64128 Len=0
283	28.333634	192.168.132.242	192.168.132.238	TCP	4444	49816	60	4444 → 49816 [ACK] Seq=17 Ack=526 Win=64128 Len=0
281	28.330733	192.168.132.242	192.168.132.238	TCP	4444	49816	60	4444 → 49816 [ACK] Seq=17 Ack=524 Win=64128 Len=0

As per the above result we identified a stable communication with multiple PUSH requests, examining the PUSH request to identify the command execution.

Conclusion

The thorough examination of the incident has allowed you to understand the sequence of events leading up to the occurrence, identifying areas for improvement in processes and procedures.

Thank You!

Hope you have learnt a lot after practically completing
Blue Team Fundamentals Exercise.

If you like the investigation & the lab scenario,
feel free to tag us & share your achievement / progress
on social media platforms :)

For Enterprise Red Team / Blue Team / Purple Team
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